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Blockchain Experiment 11

AIM: To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

Theory:
Hyperledger Fabric is an open-source enterprise-grade blockchain platform designed for building distributed applications. Unlike public blockchains, it is a **permissioned network**, meaning only authorized participants can access and perform transactions.

The architecture of Hyperledger Fabric consists of key components:

- **Peers** – Maintain ledger and run chaincode
- **Orderers** – Ensure transaction ordering
- **Certificate Authority (CA)** – Provides identities for participants
- **Channels** – Private communication layers for transactions

Chaincode (smart contracts) defines the business logic that governs how assets are created, updated, and transferred on the blockchain.

To run Fabric, tools like:

- Docker (for containerization)
- Git (for repository management)
- Node.js (for chaincode execution)
- Windows Subsystem for Linux (for Linux environment on Windows) are required.

This experiment demonstrates how to:

1. Set up a Fabric test network
2. Deploy chaincode
3. Perform asset lifecycle operations

Prerequisites

- Windows Subsystem for Linux (WSL)
- Ubuntu
- Docker
- Git
- Node.js & npm

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PROCEDURE

Step 1: Update the system

Go to https://shell.cloud.google.com and run the following commands:
sudo apt update
sudo apt upgrade -y

```
You can view your projects by running  gcloud projects list .
g2022_sonam_chhabaidiya@cloudshell:~$ sudo apt update
Get:1 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [180 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [675 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,508 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,989 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3,505 kB]
Hit:7 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease
Get:8 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:9 https://cli.github.com/packages stable InRelease [3,917 B]
Get:10 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:11 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:12 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [59.7 kB]
Get:13 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:14 https://packages.cloud.google.com/apt gcsfuse-noble/main amd64 Packages [66.1 kB]
Hit:15 http://archive.ubuntu.com/ubuntu noble InRelease
Get:16 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [1,982 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:18 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [68.7 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:20 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,640 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,153 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,377 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,692 kB]
Fetched 23.3 MB in 14s (1,645 kB/s)
```

```
g2022_sonam_chhabaidiya@cloudshell:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  libpam-systemd libsystemd-shared libsystemd0 libudev1 qemu-user qemu-user-static systemd systemd-dev systemd-sysv udev
The following packages will be upgraded:
  docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin google-cloud-cli google-cloud-cli-anthoscli
  google-cloud-cli-app-engine-go google-cloud-cli-app-engine-java google-cloud-cli-app-engine-python
  google-cloud-cli-app-engine-python-extras google-cloud-cli-bigtable-emulator google-cloud-cli-cbt google-cloud-cli-cloud-run-proxy
  google-cloud-cli-datastore-emulator google-cloud-cli-gke-gcloud-auth-plugin google-cloud-cli-istioctl google-cloud-cli-kpt
  google-cloud-cli-local-extract google-cloud-cli-minikube google-cloud-cli-nomos google-cloud-cli-package-go-module
  google-cloud-cli-pubsub-emulator google-cloud-cli-skaffold google-cloud-cli-spanner-cli kubectl libarchive13t64 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-common libpq-dev libpq5 libruby3.2 libssl-dev libssl3t64 linux-libc-dev openssl php8.3-bcmath php8.3-cgi php8.3-cli
  php8.3-common php8.3-dev php8.3-mbstring php8.3-mysql php8.3-opcache php8.3-readline php8.3-xml ruby3.2 ruby3.2-dev tzdata
49 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
Need to get 861 MB of archives.
After this operation, 25.6 MB disk space will be freed.
Get:1 https://packages.cloud.google.com/apt cloud-sdk/main amd64 google-cloud-cli-anthoscli amd64 564.0.0-0 [35.8 MB]
```

Step 2: Install Required Dependencies

```
g2022_sonam_chhabaidiya@cloudshell:~$ sudo apt install -y jq curl git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
jq is already the newest version (1.7.1-3ubuntu0.24.04.1).
curl is already the newest version (8.5.0-2ubuntu10.8).
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 10 not upgraded.
g2022_sonam_chhabaidiya@cloudshell:~$
```

Then verify everything is present:
node -v

npm -v

git --version

docker --version

docker ps

```
g2022_sonam_chhabaidiya@cloudshell:~$ node -v
v24.14.1
g2022_sonam_chhabaidiya@cloudshell:~$ npm -v
11.12.1
g2022_sonam_chhabaidiya@cloudshell:~$ git --version
git version 2.43.0
g2022_sonam_chhabaidiya@cloudshell:~$ docker --version
Docker version 29.4.0, build 9d7ad9f
g2022_sonam_chhabaidiya@cloudshell:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
g2022_sonam_chhabaidiya@cloudshell:~$
```

Step 3: Fix Docker Compose

DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
curl -SL
https://github.com/docker/compose/releases/download/v2.24.5/docker-compose
-linux-x86_64 \

-o $DOCKER_CONFIG/cli-plugins/docker-compose

chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose

docker compose version

sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true

sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF' #!/bin/bash

exec docker compose "$@"

EOF

sudo chmod +x /usr/local/bin/docker-compose docker

compose version

docker-compose version

```
g2022_sonam_chhabaidiya@cloudshell:~$ DOCKER_CONFIG=${DOCKER_CONFIG:-$HOME/.docker}
mkdir -p $DOCKER_CONFIG/cli-plugins
g2022_sonam_chhabaidiya@cloudshell:~$ curl -SL https://github.com/docker/compose/releases/dow
-o $DOCKER_CONFIG/cli-plugins/docker-compose

chmod +x $DOCKER_CONFIG/cli-plugins/docker-compose

docker compose version

sudo mv /usr/bin/docker-compose /usr/bin/docker-compose-old 2>/dev/null || true

sudo tee /usr/local/bin/docker-compose > /dev/null <<'EOF'
#!/bin/bash
exec docker compose "$@"
EOF

sudo chmod +x /usr/local/bin/docker-compose
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 58.5M  100 58.5M    0     0  37.5M      0  0:00:01  0:00:01 --:--:-- 62.1M
Docker Compose version v2.24.5
g2022_sonam_chhabaidiya@cloudshell:~$ docker compose version
Docker Compose version v2.24.5
g2022_sonam_chhabaidiya@cloudshell:~$
```

Step 4: Clone Fabric Samples

cd $HOME

git clone https://github.com/hyperledger/fabric-samples cd fabric-

samples

```
g2022_sonam_chhabaidiya@cloudshell:~$ cd $HOME
g2022_sonam_chhabaidiya@cloudshell:~$ git clone https://github.com/hyperledger/fabric-samples
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15618, done.
remote: Counting objects: 100% (221/221), done.
remote: Compressing objects: 100% (126/126), done.
remote: Total 15618 (delta 149), reused 95 (delta 95), pack-reused 15397 (from 3)
Receiving objects: 100% (15618/15618), 24.09 MiB | 15.26 MiB/s, done.
Resolving deltas: 100% (8801/8801), done.
g2022_sonam_chhabaidiya@cloudshell:~$ cd fabric-samples
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$
```

Step 5: Install Fabric Binaries and Docker Images

curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh

chmod +x install-fabric.sh

./install-fabric.sh docker samples binary

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```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ chmod +x install-fabric.sh
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

==> Already in fabric-samples repo
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.15 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
===> Will unpack to: /home/g2022_sonam_chhabaidiya/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100  120M  100  120M    0     0  18.5M      0  0:00:06  0:00:06 --:--:-- 20.6M
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
===> Will unpack to: /home/g2022_sonam_chhabaidiya/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 31.6M  100 31.6M    0     0  23.1M      0  0:00:01  0:00:01 --:--:-- 33.6M
==> Done.

Pull Hyperledger Fabric docker images
```

```
Status: Downloaded newer image for ghcr.io/hyperledger/fabric-ca:1.5.17
ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9        250MB         80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB          123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15     86dded7eda92          1GB          255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15   9972c209be47        178MB         49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15      3e25adc31a75        230MB         66.2MB
hyperledger/fabric-baseos:2.5               654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:2.5.15            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-ca:1.5                   453a0a727e82        381MB          123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82        381MB          123MB
hyperledger/fabric-ca:latest                453a0a727e82        381MB          123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:latest             86dded7eda92          1GB          255MB
hyperledger/fabric-orderer:2.5              9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:latest           9972c209be47        178MB         49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$
```

Step 6: Verify Docker images

docker images | grep fabric

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ docker images | grep fabric
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
ghcr.io/hyperledger/fabric-baseos:2.5.15    654bd4554bf9        250MB         80.4MB
ghcr.io/hyperledger/fabric-ca:1.5.17        453a0a727e82        381MB          123MB
ghcr.io/hyperledger/fabric-ccenv:2.5.15     86dded7eda92          1GB          255MB
ghcr.io/hyperledger/fabric-orderer:2.5.15   9972c209be47        178MB         49.5MB
ghcr.io/hyperledger/fabric-peer:2.5.15      3e25adc31a75        230MB         66.2MB
hyperledger/fabric-baseos:2.5               654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:2.5.15            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-baseos:latest            654bd4554bf9        250MB         80.4MB
hyperledger/fabric-ca:1.5                   453a0a727e82        381MB          123MB
hyperledger/fabric-ca:1.5.17                453a0a727e82        381MB          123MB
hyperledger/fabric-ca:latest                453a0a727e82        381MB          123MB
hyperledger/fabric-ccenv:2.5                86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:2.5.15             86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:latest             86dded7eda92          1GB          255MB
hyperledger/fabric-orderer:2.5              9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:latest           9972c209be47        178MB         49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
```

Step 7: Set Environment Variables

export PATH=$PATH:$HOME/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-samples/config peer version

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ export PATH=$PATH:$HOME/fabric-samples/bin
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ peer version
peer:
 Version: v2.5.15
 Commit SHA: 83c7930
 Go version: go1.26.0
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

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Step 8: Start the Network and Create Channel

cd $HOME/fabric-samples/test-network

./network.sh up createChannel

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ cd $HOME/fabric-samples/test-network
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ ./network.sh up createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb with c
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
/home/g2022_sonam_chhabaidiya/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] Running 3/7
 ⠿ Network fabric_test                      Created
 ⠿ Volume "compose_peer0.org1.example.com"  Created
 ⠿ Volume "compose_peer0.org2.example.com"  Created
 ⠿ Volume "compose_orderer.example.com"     Created
 ✔ Container orderer.example.com            Started
 ✔ Container peer0.org2.example.com         Started
 ✔ Container peer0.org1.example.com         Started
CONTAINER ID   IMAGE                               COMMAND             CREATED        STATUS                  PORTS
05513e7c1822   hyperledger/fabric-peer:latest      "peer node start"   1 second ago   Up Less than a second   0.0.0.0:7051->7051/t
.example.com
915aaa254ba8   hyperledger/fabric-orderer:latest   "orderer"           1 second ago   Up Less than a second   0.0.0.0:7050->7050/t
ample.com
7750c6a1214a   hyperledger/fabric-peer:latest      "peer node start"   1 second ago   Up Less than a second   0.0.0.0:9051->9051/t
.example.com
```

```
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
Status: 201
{
        "name": "mychannel",
        "url": "/participation/v1/channels/mychannel",
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-04-09 17:22:26.894 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-09 17:22:26.936 UTC 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Joining org2 peer to the channel...
Using organization 2
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-04-09 17:22:30.015 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-09 17:22:30.053 UTC 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Setting anchor peer for org1...
Using organization 1
```

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```
Generating anchor peer update transaction for Org2 on channel mychannel
++ jq '.channel_group.groups.Application.groups.Org2MSP.values += {"AnchorPeers":{"mod_policy": "Admins","value"
0"}}' /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json
++ res=0
++ configtxlator proto_encode --input /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifact
fabric-samples/test-network/channel-artifacts/original_config.pb
++ configtxlator proto_encode --input /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifact
abaidiya/fabric-samples/test-network/channel-artifacts/modified_config.pb
++ configtxlator compute_update --channel_id mychannel --original /home/g2022_sonam_chhabaidiya/fabric-samples/t
baidiya/fabric-samples/test-network/channel-artifacts/modified_config.pb --output /home/g2022_sonam_chhabaidiya/
++ configtxlator proto_decode --input /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifact
iya/fabric-samples/test-network/channel-artifacts/config_update.json
++ jq .
+++ cat /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifacts/config_update.json
++ echo '{"payload":{"header":{"channel_header":{"channel_id":"mychannel", "type":2}},"data":{"config_update":{'
ps":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '"groups":' '{},' '"mod_policy":' '"",' '"polici
0"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '
"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '}' '},' '"values":' '{' '"MSP":' '{' '"mod_policy":'
mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"po
oups":' '{' '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '"groups":' '{},' '"mod_policy":' '"Admins",'
on":' '"0"' '},' '"Endorsement":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers"
:' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '}' '},' '"values":' '{' '"AnchorPeers":' '{'
peer0.org2.example.com",' '"port":' 9051 '}' ']' '},' '"version":' '"0"' '},' '"MSP":' '{' '"mod_policy":' '"",'
olicy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policie
++ configtxlator proto_encode --input /home/g2022_sonam_chhabaidiya/fabric-samples/test-network/channel-artifact
m_chhabaidiya/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.tx
2026-04-09 17:22:30.712 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialize
2026-04-09 17:22:30.727 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 9: Deploy Chaincode

./network.sh deployCC \

-ccn basic \

-ccp ../asset-transfer-basic/chaincode-javascript \

-ccl javascript

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples$ cd test-network

./network.sh deployCC \
  -ccn basic \
  -ccp ../asset-transfer-basic/chaincode-javascript \
  -ccl javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ '[' false = true ']'
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[].package_id)'
```

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```
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel '
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.exampl
erOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic
st-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.p
test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert
+ res=0
2026-04-09 17:29:16.170 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [4a32a0adf5b927408e350c
2026-04-09 17:29:16.170 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [4a32a0adf5b927408e350c
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1M
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1M
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
```

Step 10: Set Peer Environment Variables

export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"

export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt

export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp

export CORE_PEER_ADDRESS=localhost:7051

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-samples/config
export PATH=$PATH:$HOME/fabric-samples/bin

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"

export CORE_PEER_TLS_ROOTCERT_FILE=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/

export CORE_PEER_MSPCONFIGPATH=$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/user

export CORE_PEER_ADDRESS=localhost:7051
```

Step 11: Initialize Ledger

peer chaincode invoke \

-o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls \

--cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert. pem \

-C mychannel \

-n basic \

--peerAddresses localhost:7051 \

--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \

--peerAddresses localhost:9051 \

--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \

-c '{"function":"InitLedger","Args":[]}'

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tl
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/c
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/c
  -c '{"function":"InitLedger","Args":[]}'
2026-04-09 17:37:00.083 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 12: Query All Assets

peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAs
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"},{"AppraisedValue":400,"Color":"red
ppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"},{"AppraisedValue":600,"Color":"yell
"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"},{"AppraisedValue":800,"Color":"wh
"}]
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 13: Read a Specific Asset

peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset5"]}'

```
"}]
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset
{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 14: Create a New Asset

peer chaincode invoke \

-o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls \

--cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert. pem \

-C mychannel \

-n basic \

--peerAddresses localhost:7051 \

--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \

--peerAddresses localhost:9051 \

--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \

-c '{"function":"CreateAsset","Args":["asset7","blue","10","Anuprita","500"]} '

```
bash: 200 peer: command not found
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"CreateAsset","Args":["asset7","blue","10","Sonam","500"]}'
2026-04-09 17:41:42.358 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"{\"ID\":\"as
\":\"Sonam\",\"AppraisedValue\":500}"
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 15: Transfer Asset

peer chaincode invoke \

-o localhost:7050 \

--ordererTLSHostnameOverride orderer.example.com \

--tls \

--cafile
$HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert. pem \

-C mychannel \

-n basic \

--peerAddresses localhost:7051 \

--tlsRootCertFiles
$HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ peer chaincode invoke \
  -o localhost:7050 \
  --ordererTLSHostnameOverride orderer.example.com \
  --tls \
  --cafile $HOME/fabric-samples/test-network/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-ce
  -C mychannel \
  -n basic \
  --peerAddresses localhost:7051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt \
  --peerAddresses localhost:9051 \
  --tlsRootCertFiles $HOME/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/peers/peer0.org2.example.com/tls/ca.crt \
  -c '{"function":"TransferAsset","Args":["asset7","Sneha"]}'
2026-04-09 17:43:47.043 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Sonam"
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

Step 16: Stop the Network

cd $HOME/fabric-samples/test-network

./network.sh down

```
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$ cd $HOME/fabric-samples/test-network
./network.sh down
Using docker and docker-compose
Stopping network
project name can't be empty. Use `--project-name` to set a valid name
Error response from daemon: get docker_orderer.example.com: no such volume
Error response from daemon: get docker_peer0.org1.example.com: no such volume
Error response from daemon: get docker_peer0.org2.example.com: no such volume
Removing remaining containers
05513e7c1822
915aaa254ba8
7750c6a1214a
d42ffab8f800
378f6ce27499
Removing generated chaincode docker images
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f5
Deleted: sha256:fee56cedb019bbf6187188cc539b007771e1e9ea1ede2ceb04a08b0aa01d5a89
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbe664c87b24c035bee15cbcc7f5
Deleted: sha256:81915028ae76fab712f5abc634afe41939728cf1bd4c17f4e3facdd016acef46
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
481282afbc43: Pull complete
aecba0016ef2: Download complete
Digest: sha256:1487d0af5f52b4ba31c7e465126ee2123fe3f2305d638e7827681e7cf6c83d5e
Status: Downloaded newer image for busybox:latest
g2022_sonam_chhabaidiya@cloudshell:~/fabric-samples/test-network$
```

CONCLUSION:

The experiment successfully demonstrated the setup and implementation of a Hyperledger Fabric blockchain network. The network was initialized, and smart contract (chaincode) was deployed on the channel. Various asset operations such as initialization, querying, creation, and transfer were performed using chaincode functions.

This experiment helped in understanding how a permissioned blockchain system works, where only authorized participants can interact with the network. It also highlighted the importance of blockchain features like security, transparency, and immutability in managing digital assets efficiently.